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In [7]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_csv("BostonHousing.csv")

X = df.drop('medv', axis=1)
y = df['medv']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

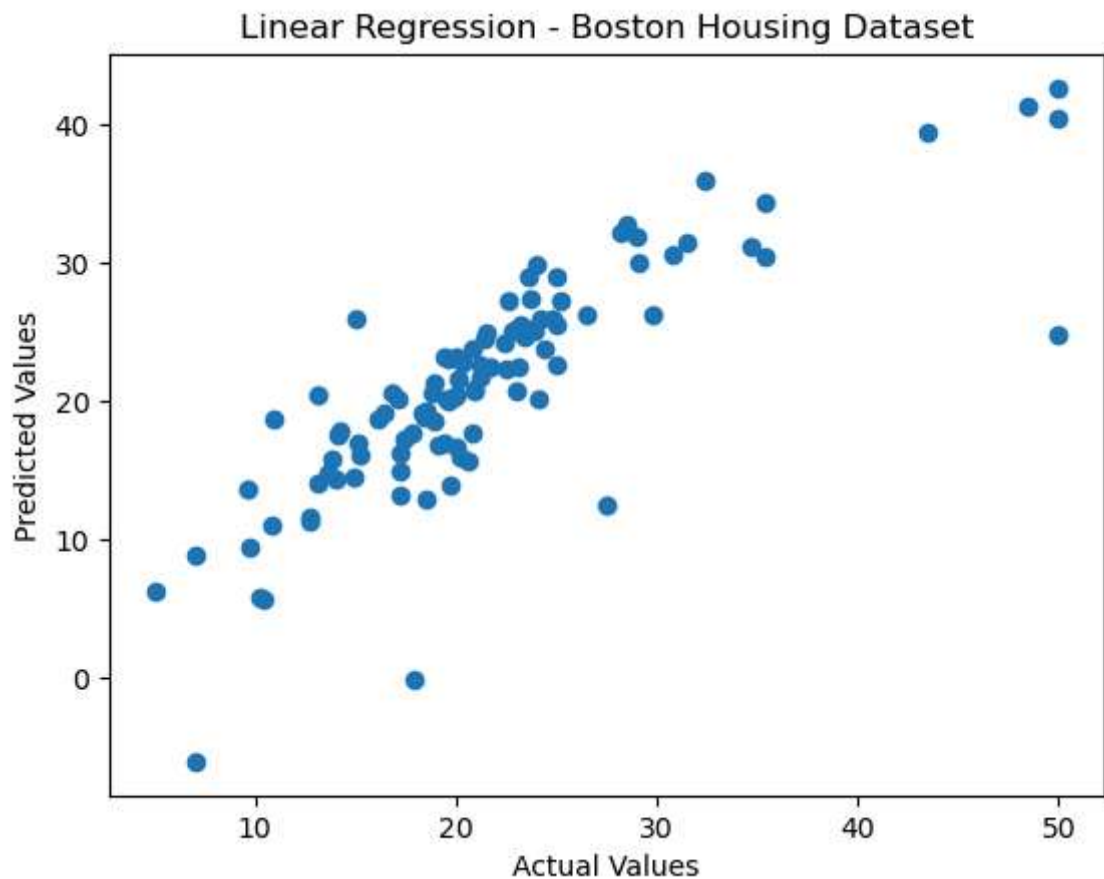
model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

plt.scatter(y_test, y_pred)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Linear Regression - Boston Housing Dataset")
plt.show()

print("Linear Regression - Boston Housing Dataset")
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R^2 Score:", r2_score(y_test, y_pred))

```



Linear Regression - Boston Housing Dataset

Mean Squared Error: 24.291119474973655

R<sup>2</sup> Score: 0.6687594935356301

```
In [10]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/mpg.csv"
df = pd.read_csv("auto-mpg.csv")

df = df.dropna()

X = df.drop('mpg', axis=1)
y = df['mpg']

X = pd.get_dummies(X, drop_first=True)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

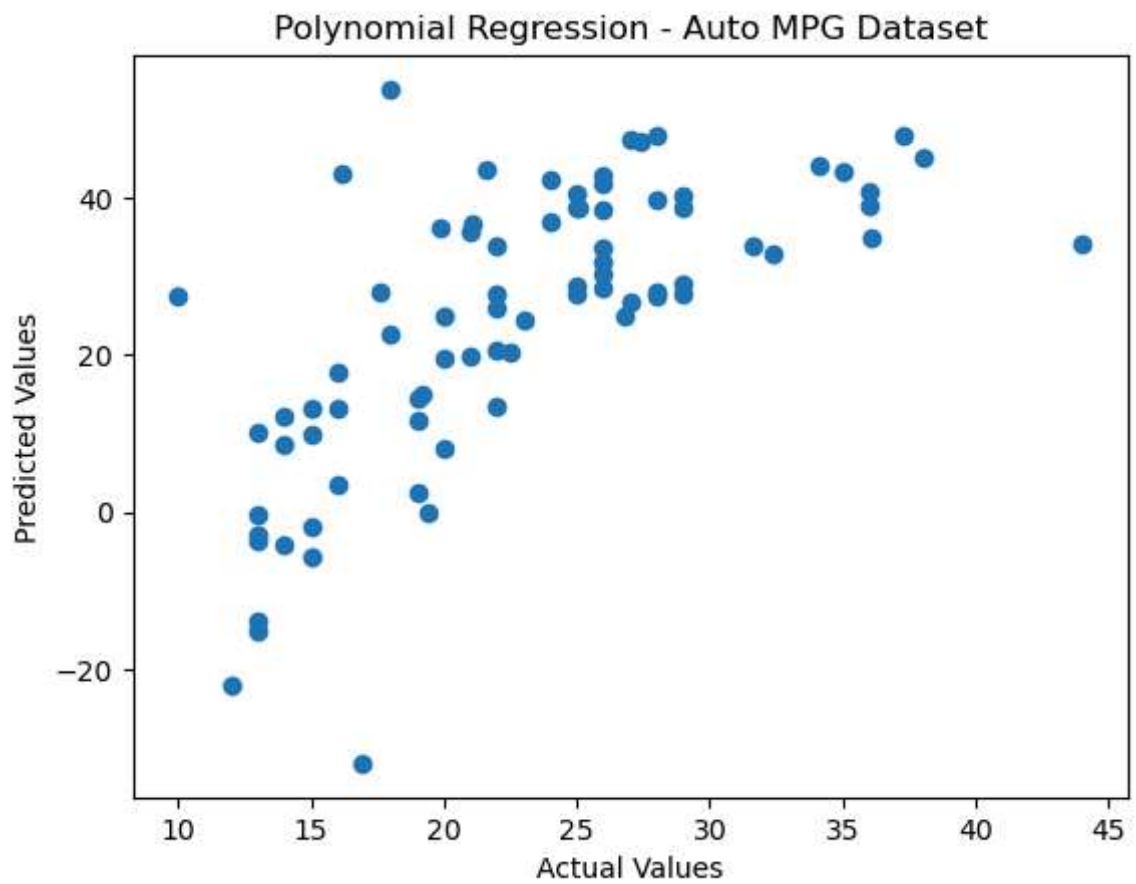
poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

model = LinearRegression()
model.fit(X_train_poly, y_train)

y_pred = model.predict(X_test_poly)

plt.scatter(y_test, y_pred)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Polynomial Regression - Auto MPG Dataset")
plt.show()

print("Polynomial Regression - Auto MPG Dataset")
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R^2 Score:", r2_score(y_test, y_pred))
```



Polynomial Regression - Auto MPG Dataset

Mean Squared Error: 200.50117419668598

R<sup>2</sup> Score: -2.9282696526885283